SVM for Hand Based Identification

Classifier for identification based on palmar and dorsal images of hands

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Abstract

Biometric hand recognition is an authentication technique increasingly applied in security and personal identification. This project focuses on developing and evaluating a recognition system based on hand images (palm and dorsal) using Support Vector Machine (SVM) classifiers. Unlike previous work that employed convolutional neural networks (CNNs) for automatic feature extraction to determine gender, this approach utilizes handcrafted features, specifically Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), combined with SVM for classification. The report details the complete workflow, from data preprocessing and feature extraction to system implementation and performance evaluation using standard biometric metrics. The ultimate objective is to create an automatic supervised hand recognition system.

1 Task description

Biometric hand recognition is an increasingly adopted technique for authentication and personal identification. This project focuses on developing a recognition system based on images of both the palm and dorsal side of the hand, aiming to accurately classify individuals. While previous approaches have relied on convolutional neural networks (CNNs) for automatic feature extraction, which require large datasets and significant computational resources, this project adopts handcrafted feature extraction techniques—specifically Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG)—combined with a Support Vector Machine (SVM) classifier. The system workflow includes data preprocessing, feature extraction, model training, and performance evaluation using standard biometric metrics. The main challenge is to achieve high recognition accuracy using interpretable, handcrafted features, ultimately creating a robust, automatic supervised hand recognition system suitable for security and personal identification contexts.

1.1 Real-world applications

Biometric hand recognition systems have a wide range of practical applications in security, identification, and access control. In security-sensitive environments such as airports, banks, or government facilities, hand-based authentication can serve as an additional or alternative method to passwords and ID cards, reducing the risk of unauthorized access. In personal identification, these systems can be integrated into smartphones, laptops, and smart home devices to provide secure, convenient, and contactless user verification. Beyond access control, hand biometrics can be employed in forensic investigations to identify individuals from images or partial hand prints, as well as in healthcare settings for patient identification and record management. The use of both palm and dorsal hand features, combined with reliable feature extraction techniques such as LBP and HOG, ensures that recognition systems can operate effectively under varying imaging conditions, enhancing both accuracy and robustness in real-world scenarios

2 Related work

Hand-based biometric recognition is a reliable approach for authentication, leveraging unique features of the palm, dorsal hand, and hand geometry. A key study is "11K Hands: Gender recognition and biometric identification using a large dataset of hand images" [1], which demonstrated that CNN-extracted features combined with SVM classifiers can accurately identify individuals using standard RGB images. This work highlights that simple hand images can provide sufficient discriminatory information, motivating feature-based approaches.

"Robust Pose Invariant Shape and Texture based Hand Recognition" [4] showed that combining shape and texture descriptors from segmented hand images achieves high accuracy without deep learning. Surveys such as "Palmprint Recognition: Extensive Exploration of Databases, Methodologies, Comparative Assessment, and Future Directions" [2] confirm that descriptors like Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) are robust, computationally efficient, and interpretable.

Recent hybrid studies, such as "Recognition of Dorsal Hand Vein in Small-Scale Sample Database Based on Fusion of ResNet and HOG Feature" [3] , demonstrate that combining shallow descriptors with deep features can enhance robustness.

These works support our approach of using LBP

and HOG descriptors from both palm and dorsal images with an SVM classifier, providing a computationally efficient and interpretable system suitable for real-world biometric applications.

3 Dataset Description

The dataset used for this project is a public dataset called **11K Hands**, containing hand images from various subjects, each accompanied by associated metadata. It includes a total of 11,076 hand images with a resolution of 1600x1200 pixels. The images were captured from a base of 190 subjects ranging in age from 18 to 75 years old. The dataset features photos of both sides of the hand (palm and dorsal) and provides multiple images for each subject.

The available metadata for each picture include:

- Subject ID
- Subject gender
- Subject age
- Hand skin color
- Hand side
- Right or left hand
- Presence of accessories or nail polish

Such metadata will be essential for image sampling, in the CNN training phase, and during the testing phase to evaluate the correctness of the predictions obtained.

3.1 Metadata Classification

The following classifications can be made using the available metadata:

Hand side	Left	Right
Dorsal	2786	2895
Palmar	2585	2810

Table 1: Division by hand side

Presence of accessories	Number
No	7865
Yes	3211

Table 2: Division by presence of accessories

Skin color	Number
Dark	758
Fair	3493
Medium	6495
Very fair	330

Table 3: Division by skin color

From the classifications reported above, the dataset appears to be balanced regarding the global subdivision based on hand side (palm/back) and left/right hand.

Gender	Number
Female	7109
Male	3967

Table 4: Division by gender

3.2 Subject Distribution

The main issue concerns the subdivision of images by subject and by gender, and the distribution of hand sides (palm/dorsal) within individual samples. Dividing the images by subject reveals a non-homogeneous distribution:

Subject group	Subject's number %	Associated images %
Top 19	10.00%	20.93%
Top 30	15.79%	30.96%
Bottom 30	15.79%	6.62%

Table 5: Subject distribution by number of associated images

Analyzing the images for each subject in detail, we find a non-homogeneous subdivision regarding the number of images per hand side:

Minimum threshold	Number of subjects	Percentage of subjects
40%	33	34.78%
50%	178	93.68%

Table 6: Distribution of subjects by minimum threshold per hand side

3.3 Data sampling

In addition to the issues mentioned earlier, the sampling phase was designed to minimize bias. Specifically, images containing accessories were excluded from the training phase to prevent their influence. Furthermore, to ensure the integrity of the evaluation process, a policy was implemented to avoid using any images from the training set in the test set.

The sampling strategy incorporates elements from two key techniques:

- **Random sampling**, where each item in the dataset has an equal chance of being selected, ensuring an unbiased sample
- **Oversampling and undersampling**, which are applied to address class imbalances by either adding or removing samples from certain classes

Now, let's take a closer look at the sampling pipeline. After scanning and loading the metadata file containing information about the images and subjects in the dataset, we constructed a custom data structure—a Python dictionary—that enables efficient manipulation and selection of data. Given the specific criteria for each selection, the sampling phase plays a crucial role in shaping the model's performance.

3.4 Distribution of Images Across Training and Closed Test Splits

Initially, we randomly select n subjects, where the number of subjects corresponds to the value of the "num_sub" variable. In the case of closed-set experiments, these subjects will be used to extract images for both the training and test sets. For each selected subject, we then randomly sample n palm/dorsal image pairs and store them in a data structure that also records the individual's identity, indicated by the "person_id" label.

The number of images to extract will be calculated as follows:

- 70% of the value passed into the "num_img" variable (rounded down) dedicated to the training phase
- 30% of the value passed in the "num_img" variable (rounded down) dedicated to the test phase

Example
num subjects = 10 and num_img = 100

- We will extract 70 pairs of images (palm and dorsal) for each subject, to be dedicated to training, with possible repetitions
- We will extract 30 pairs of images (palm and back) for each subject, to be dedicated to the test, with possible repetitions, including the possibility of fishing for some of the images collected in the training

3.5 Subdivision of images between training and open test sets

When performing recognition in open-set mode, the procedures for selecting subjects and images must be adjusted. During the testing phase, a number of n impostor subjects are randomly chosen from among those initially extracted for the gallery. To mark a subject as an impostor—an essential step for the subsequent evaluation of the biometric system—their identity label is set to -1.

4 Image preprocessing

Hand image preprocessing is a crucial step in biometric identification systems. It improves data quality and reduces irrelevant variations caused by lighting, hand orientation, position, or complex backgrounds. Operations such as resizing, contrast adjustment, noise removal, and normalization allow the model to focus on distinctive hand features, including skin texture, veins, palm lines, and overall shape. Overall, properly prepared images make the system more robust and reliable, enhancing recognition accuracy and the ability to distinguish legitimate users from impostors. Without effective preprocessing, the model's performance could be compromised by irrelevant variability.

4.1 Single image preprocessing

Before being passed to the functions that deal with feature extraction, the images were subjected to specific preprocessing, which include:

- **CustomLBPCannyTransform**: It is applied to all palm images that must be subjected to the feature extraction process through LBP, in sequence it executes:
  - Resizing the image to a standard dimension of 150×150 pixels for analysis
  - Converts the image to grayscale
  - Increases the contrast
  - Applies the Canny filter to reduce noise and highlight the most important features
- **CustomLBPTransform**: This process is applied to all dorsal images used for feature extraction via LBP. It follows the same steps as described above, except that the Canny filter is not applied
- **CustomHOGTransform**: It is applied to all images of both the dorsal and the palm that must be subjected to the feature extraction process through HOG, in sequence it executes:
  - Converts the image to RGB
  - Applies a Gaussian filter to the image to reduce noise
  - Resizing the images to standard dimensions of 1024×1024 pixels for dorsal images and 150×150 pixels for palm images prior to analysis

5 System implementation

With an understanding of the image preprocessing steps, we can now turn to the implementation of the biometric recognition system. The system follows a structured pipeline, progressing from image acquisition and preprocessing to final classification using Support Vector Machines (SVM)

5.1 Biometric Recognition Pipeline

Setting aside the preprocessing already discussed, the system can be organized into the following key steps:

- Feature extraction performed using LBP and HOG
- Training of the SVM Model
- Fusion of predictions
- Performance Evaluation

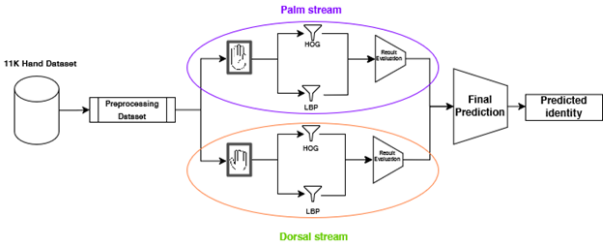

Figure 1: Diagram of the implemented pipeline in the biometric recognition system

5.2 Feature extraction

In the previous CNN-based project, features were automatically learned by the neural network during training. In contrast, for the current project we opted to use Support Vector Machines (SVM), a classifier that cannot extract features directly from raw images. Therefore, feature extraction was performed using two well-established techniques: Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG). These methods produce numerical representations of the images, which are then fed to the SVM classifier to distinguish the subjects

5.2.1 Local Binary Pattern (LBP)

Local Binary Pattern (LBP) is a technique used to describe the texture of an image based on the intensity values of surrounding pixels. The method works as follows:

- The image is divided into cells of fixed size (in our case, 3×3 pixels)
- For each pixel, its intensity value is compared with those of its neighbors:
  - If a neighboring pixel has a value greater than or equal to the central pixel, it is assigned a 1
  - If it is smaller, it is assigned a 0
- The resulting binary pattern is converted into a decimal number
- A histogram of LBP values is constructed to describe the distribution of textures within the image

LBP is particularly effective for texture recognition because it is robust to changes in lighting and captures distinctive details of skin patterns and palm lines. In our project, LBP was used to extract local information about the skin texture and palm lines. For each pixel, the intensity is compared with surrounding pixels to generate a binary map, which is then transformed into frequency histograms. These histograms provide a compact and informative representation of the image, suitable for further analysis by the classifier.

5.2.2 Histogram of Oriented Gradients (HOG)

The Histogram of Oriented Gradients (HOG) is a feature extraction technique that analyzes the distribution of intensity gradients in an image, effectively capturing the shapes and contours of objects. The method proceeds as follows:

- Gradients of pixel intensities are computed to measure intensity variations between adjacent pixels
- The image is divided into cells (4×4 pixels in our case)
- For each cell, the dominant direction of the gradients is determined
- A histogram of these predominant directions is created for each cell
- Histograms from all cells are combined to form a feature vector representing the entire image

HOG is widely used in object and human recognition because it captures the global shape of a hand more effectively than pixel-based methods alone. In our project, HOG is applied to both dorsal and palm images to extract patterns corresponding to the grooves and lines of the hand, providing a robust representation for the classifier.

5.2.3 Feature extraction combination

LBP analyzes hand texture to capture fine details, while HOG detects overall shape and contours. Combined, they provide a complementary representation that enhances the accuracy and robustness of biometric analysis.

6 Support Vector Machine classification

The extracted features, as described above, serve as the foundation for implementing Support Vector Machine (SVM) classifiers. SVMs are supervised machine learning models that are particularly effective when dealing with high-dimensional features and relatively small training datasets. One of their main advantages is robustness to overfitting, achieved through regularization techniques. Compared to CNNs, SVMs are also computationally lighter during both training and inference, making them well-suited for our project.

The primary goal of an SVM is to identify an optimal hyperplane that best separates data from different classes. To achieve this, the model:

- Transforms the data into a higher-dimensional space if necessary, using kernel functions
- Finds the hyperplane with the maximum margin between classes—the greatest distance between the closest points of different classes, known as support vectors
- Assigns new instances to the corresponding class based on their position relative to this optimal hyperplane

6.1 Fusion prediction

In our biometric recognition system, we use images of both the palm and the dorsal side of the hand to improve classification accuracy. The idea behind prediction fusion is to combine information from both views to achieve more robust and reliable recognition. The palm provides clear details on lines and skin texture, while the dorsal side offers information on hand shape and vein distribution. Additionally, some features may be subtle in one view but more pronounced in the other, and fusion helps reduce errors caused by environmental or capture-phase variations.

After extracting features from both views, we train four separate SVM models, grouped by hand side. Specifically, for each side, one model is trained with LBP features and another with HOG features, covering both the palm and dorsal views.

Each model produces a classification probability, or score, for each subject. These scores are then combined using a weighted fusion strategy, allowing the most reliable views to contribute more heavily to the final decision. By integrating information from both the palm and dorsal views, this approach increases the system's reliability, compensates for potential deficiencies in either view, and enhances the overall accuracy of the biometric recognition.

7 Performance evaluation

To assess the effectiveness of the proposed models, we conduct a comprehensive performance evaluation using standard metrics. This section presents the experimental setup, evaluation criteria, and the results obtained for each model. The goal is to quantitatively measure the accuracy, robustness, and generalization ability of the classifiers across different hand views and feature extraction methods.

7.1 Evaluation metrics used

After obtaining the combined prediction through the fusion of palm and back probabilities, we evaluated the effectiveness of the system using several key metrics to measure recognition performance. More specifically, the metrics adopted are:

- **Accuracy**: Percentage of correct identifications out of the total tests performed. Calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  Where:
  TP (True Positive) = Correct identifications of authorized users
  TN (True Negative) = Correct rejections of unauthorized users
  FP (False Positive) = Misidentifications of unauthorized users
  FN (False Negative) = Incorrect rejections of authorized users

- **Confusion Matrix**: It provides a detailed view of the model's performance by showing how predictions are distributed between correct and incorrect classes, allowing the identification of the main causes of recognition errors

- **False Acceptance Rate (FAR):** Indicates the probability that the system will erroneously accept an unauthorized user (a false positive), which is particularly important in security contexts where a high FAR can lead to unauthorized intrusions. Calculated as:

$$\text{FAR} = \frac{FP}{FP + TN}$$

- **False Rejection Rate (FRR):** Indicates the probability that the system will erroneously reject an authorized user (a false negative), which can compromise usability by making it difficult for authorized users to access the system. Calculated as:

$$\text{FRR} = \frac{FN}{FN + TP}$$

8 Result and performance analysis

The identification model supports both open-set and closed-set scenarios. In closed-set identification, all individuals are assumed to be in the database, and the system always returns a match. In open-set identification, the system must decide whether to accept or reject unknown individuals. While closed-set systems risk misidentification, open-set systems must manage similarity thresholds to avoid recognizing individuals not in the database.

8.1 Closed Set Identification result

The first metric analyzed in closed-set identification is the confusion matrix, which helps identify imbalances in the dataset, such as identities that are predicted multiple times. Below the confusion matrix obtained from a set of 40 subjects and 25 images:

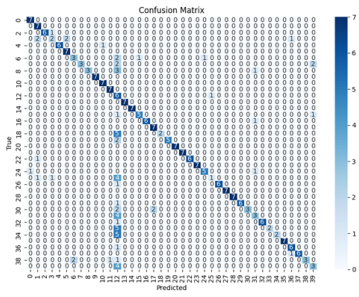

Figure 2: Confusion matrix of the HOG dorsal prediction

It is worth noting that, despite a relatively homogeneous diagonal (indicating a high number of correct recognitions), one identity, number 8, is predicted significantly more often than the others—59 times in total.

This suggests a strong bias, likely due to an imbalance in the dataset.

The second metric in closed-set identification is the CMC curve, which shows the cumulative probability that the correct identity of an individual appears among the first k matches returned by the biometric system. In our case, the CMC curve was computed for each of the four identifications performed using SVC, as well as for the final multibiometric recognition. The results are presented below:

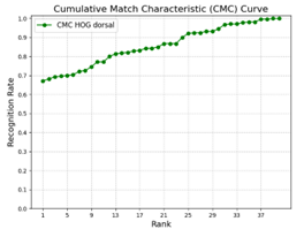

Figure 3: Confusion matrix of the HOG dorsal prediction

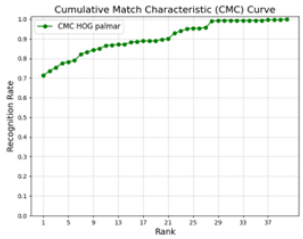

Figure 4: CMC graph of the HOG palmar prediction

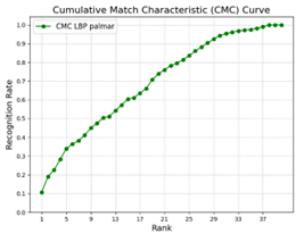

Figure 5: CMC graph of the LBP palmar prediction

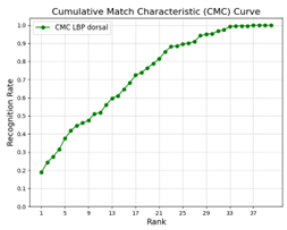

Figure 6: CMC graph of the LBP dorsal prediction

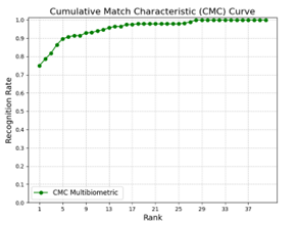

Figure 7: CMC graph of the cumulative prediction

The CMC curve reveals that identification is generally inaccurate for most test cases. While the system performs well when using HOG features on both the palm and dorsal sides of the hand (with a rank-5 accuracy of nearly 80% in the first case), performance significantly drops in other cases. The worst performance occurs with LBP features on the palm, where the model starts with only 10% accuracy and increases very slowly, reaching 100% correct identifications only at rank 38. On a more positive note, the system shows good performance in multi-biometric recognition, achieving 90% accuracy at rank 10.
Finally, the last metric used to evaluate the biometric system is accuracy, calculated for each of the four test cases and the final multi-biometric prediction. The results are summarized in the table below:

Test Case	Accuracy
LBP Palmar	10.2%
LBP Dorsal	18.7%
HOG Palmar	71.8%
HOG Dorsal	67.3%
Multi Biometric System	75.0%

Table 7: Accuracy results for the various test cases and the final multi-biometric prediction.

These results clearly demonstrate the effectiveness of multi-biometric identification, achieving 75% accuracy despite the relatively poor performance of some of the individual models.

8.2 Open Set Identification

The performance of the open set recognition system was evaluated using FAR, FRR, and EER (Equal Error Rate). The combined graphs illustrate the relationship between FAR and FRR as the acceptance threshold varies, with the EER representing the point of balance between false alarms and false rejections.

From the graphs, the FAR curves generally show sharp drops, indicating that the model moves abruptly from a very permissive to a highly restrictive state. This behavior may be caused by balance issues or a strong reliance on specific features. The FRR curves rise quickly and then level off, reflecting a sudden transition from a state in which nearly all legitimate users are accepted to one in which almost none are. This pattern may point to limited model generalization or an inadequately calibrated distribution of similarity scores.

This abrupt behavior is less pronounced in the multi-biometric curves, suggesting that combining multiple biometric modalities increases the system's stability and robustness.

Looking at the EER, LBP features result in an average of about 0.5, indicating poor discrimination between genuine users and impostors. HOG features achieve an average EER of 0.04, demonstrating effective class separation. Multi-biometric recognition shows an intermediate EER of 0.314, improving upon LBP but not reaching the high performance of HOG.

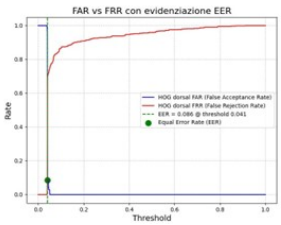

Figure 8: FAR and FRR curves with EER for dorsal HOG

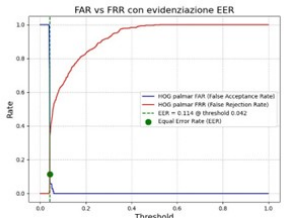

Figure 9: FAR and FRR curves with EER for palmar HOG

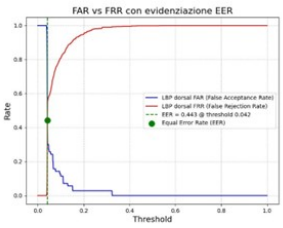

Figure 10: FAR and FRR curves with EER for dorsal LBP

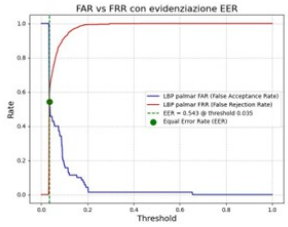

Figure 11: FAR and FRR curves with EER for palmar LBP

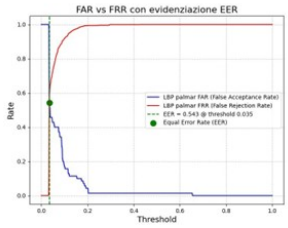

Figure 12: FAR and FRR curves with EER for multibiometric fusion

Finally we can measure the accuracy between the various models:

Test Case	Accuracy
LBP Palmar	13,5%
LBP Dorsal	21,2%
HOG Palmar	66,1%
HOG Dorsal	31,3%
Multi Biometric System	68,9%

Table 8: Accuracy results for the various test cases and the final multi-biometric prediction.

9 Possible Improvements and Future Developments

Significant improvements could be achieved by optimizing SVM parameters, such as experimenting with different kernels to enhance data separability. Feature selection techniques could reduce dimensionality and improve computational efficiency. The inclusion of a soft biometric trait, like gender detected via CNN, could provide a secondary verification threshold, enhancing recognition accuracy without being used for identification. A hybrid approach combining manual feature extraction and neural networks could offer more structured input to CNNs, improving accuracy while limiting dataset size and computational load.

10 Conclusions

The project demonstrated that a biometric recognition system based on hand-crafted features (LBP and HOG) combined with an SVM classifier can potentially provide acceptable performance, with advantages in terms of interpretability and reduced computational complexity. However, it has been widely highlighted that the suggested implementation suffers from a very serious dataset balancing problem which negatively impacts the quality of biometric recognition. It was then demonstrated how a multi-biometric recognition system behaves in a decidedly superior manner compared to a single biometric trait system, constantly managing to outperform the best single biometric trait system even in cases of terrible performance of the other models on which it is based. The developed system could potentially be used in several applications, including:

- Access control in secure environments
- Biometrics-based authentication systems
- Automated identification in forensics

References

[1] 11k hands: Gender recognition and biometric identification using a large dataset of hand images. 1

[2] Palmprint recognition: Extensive exploration of databases, methodologies, comparative assessment, and future directions. 1

[3] Recognition of dorsal hand vein in small-scale sample database based on fusion of resnet and hog feature. 1

[4] Robust pose invariant shape and texture based hand recognition. 1